

Toward a Terahertz-Clocked Half Adder Using Graphene Plasmons as Bits: Conceptual Foundations, Architecture, and a Quantitative Roadmap

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Abstract

We present a conceptual model of a terahertz-clocked half adder in which the logical state is carried by graphene surface plasmons rather than by electrical currents. The architecture builds on a six-year arc of prior work by the author: a 2019 proposal for field-coupled classical logic gates [1], a 2024 conceptual design of a plasmon-generated programmable logic circuit using AA/AB-stacked trilaminar graphene [2], and the 2023 experimental demonstration of on-chip terahertz launching of ultrashort graphene plasmon wavepackets by Yoshioka et al. [3]. The device consists of a dual-gated bilayer-graphene channel in which the upper layer carries signal plasmons and the lower layer, biased to generate π -shifted plasmons, performs NOT logic by destructive interference at designated interaction nodes. We fix a consistent terminology in which the *bit* is identified with a plasmon wavepacket of fixed amplitude, present (“1”) or absent (“0”) at a designated read-out cell. We verify the design against the complete four-row half-adder truth table, give first-order quantitative estimates of the propagation length ($L_p \sim 1\text{--}3\ \mu\text{m}$ at 10 THz), gate delay ($\tau_g \sim 0.1\ \text{ps}$), and intrinsic energy per bit ($E_{\text{bit}} \sim 10^{-18}\ \text{J}$, exclusive of antenna and read-out conversion overheads), and quantify the phase-tolerance budget required for reliable NOT operation ($|\Delta\phi| < \pi/12$ for $> 90\%$ extinction). A dedicated Limitations section enumerates the open problems — cryogenic operation, cascading, integrated read-out, phase coherence at scale, and fabrication uniformity — that bound the current proposal.

Authorship and Status Disclaimer

This manuscript is derived from the author’s original papers [1] and [2]. Generation of content involved the use of Anthropic’s Claude (commercial tier) at the author’s direction, followed by revisions incorporating reviewer comments sourced from Microsoft Copilot. The author wishes to declare that they lack formal expertise in lattice cryptography or protocol-level security analysis, and that the document has not received independent expert review. Consequently, the construction and security claims presented herein must be treated solely as a starting point for discussion and not as peer-reviewed cryptographic guidance. Formal expert review is required prior to any practical implementation.

1. Introduction

After half a century of CMOS scaling, classical computing is approaching physical limits in feature size, switching energy, and clock frequency. Quantum computing has dominated the discussion of post-CMOS architectures, but it solves a different class of problems and does not relieve the energy burden of *classical* arithmetic that still underlies machine learning, scientific simulation, and edge inference. This motivates the search for alternative classical substrates that can offer terahertz-scale clocks at lower energy per operation than scaled CMOS [4, 5].

Surface plasmons in graphene are a natural candidate. They confine electromagnetic energy to nanometre-scale volumes, propagate at ten-to-hundred terahertz frequencies, and can be modulated electrostatically by gating the underlying graphene through its charge-neutrality point (CNP) [6–8]. The practical bottleneck has historically been launching well-defined plasmon wavepackets on-chip, which Yoshioka et al. [3] addressed in 2023 by integrating a CMOS-compatible terahertz emitter with a graphene channel.

The contribution of the present paper is to assemble these ingredients into a complete arithmetic primitive — a half adder — and to specify the operation of the device in enough detail that it can be quantitatively evaluated,

simulated, or fabricated. The paper proceeds bottom-up: Section 2 establishes the conceptual foundations — field-coupled logic, the unifying Combo gate of [1], graphene plasmons as switchable bits, and on-chip terahertz launching [3]; Section 3 introduces the device architecture; Section 4 maps the half-adder logic onto the gate substrate and verifies it against the full truth table; Section 5 supplies the quantitative figures of merit; and Section 6 enumerates the limitations of the present proposal that bound an experimental realisation.

2. Conceptual foundations

2.1 From field-coupled logic to plasmonic bits

The earliest version of this programme, presented in 2019 [1], proposed that classical logic could be implemented purely by spatially arranged electric fields rather than by current flow. In that model the OR gate and the AND gate are not separate physical objects but the *same* structure with the output placed at different distances from the inputs (Fig. 1). When the output is close, either input is sufficient to drive it (OR); when the output is farther, only the superposition of two input fields exceeds the threshold (AND). The NOT gate is realised separately by a polarity-inverting interaction such as exciton trapping at the output edge.

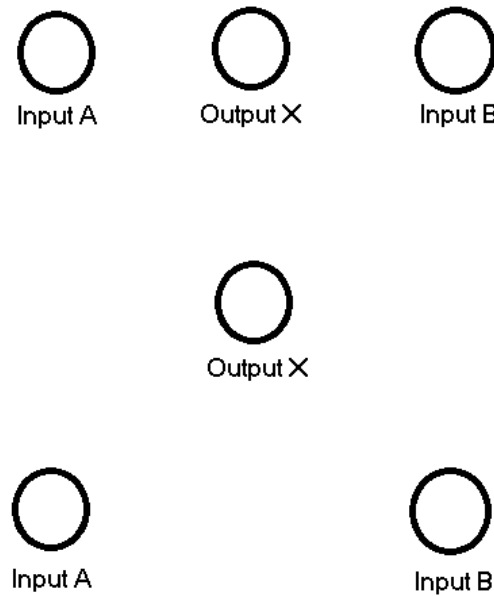


Fig. 1. Original 2019 proposal: OR (top, output close to the inputs) and AND (bottom, output displaced) implemented by the same field-coupling primitive. Reproduced from [1].

Replacing the static electric field of [1] with a propagating plasmon wavepacket changes very little at the conceptual level — the bit is still an electric-field configuration — but it gives the bit a finite group velocity, a well-defined phase, and a very short temporal extent. These three properties are what convert a static gate into a clocked, high-frequency logic device.

2.2 The Combo gate

The unifying construct introduced in [1] is the *Combo gate*: a single physical region with two output cells at different distances from the input pair, so that the AND output and the OR output are produced simultaneously. This is the structure that the half adder of Section 4 inherits, since Sum and Carry are themselves an OR-with-cancellation and an AND of the same two inputs.

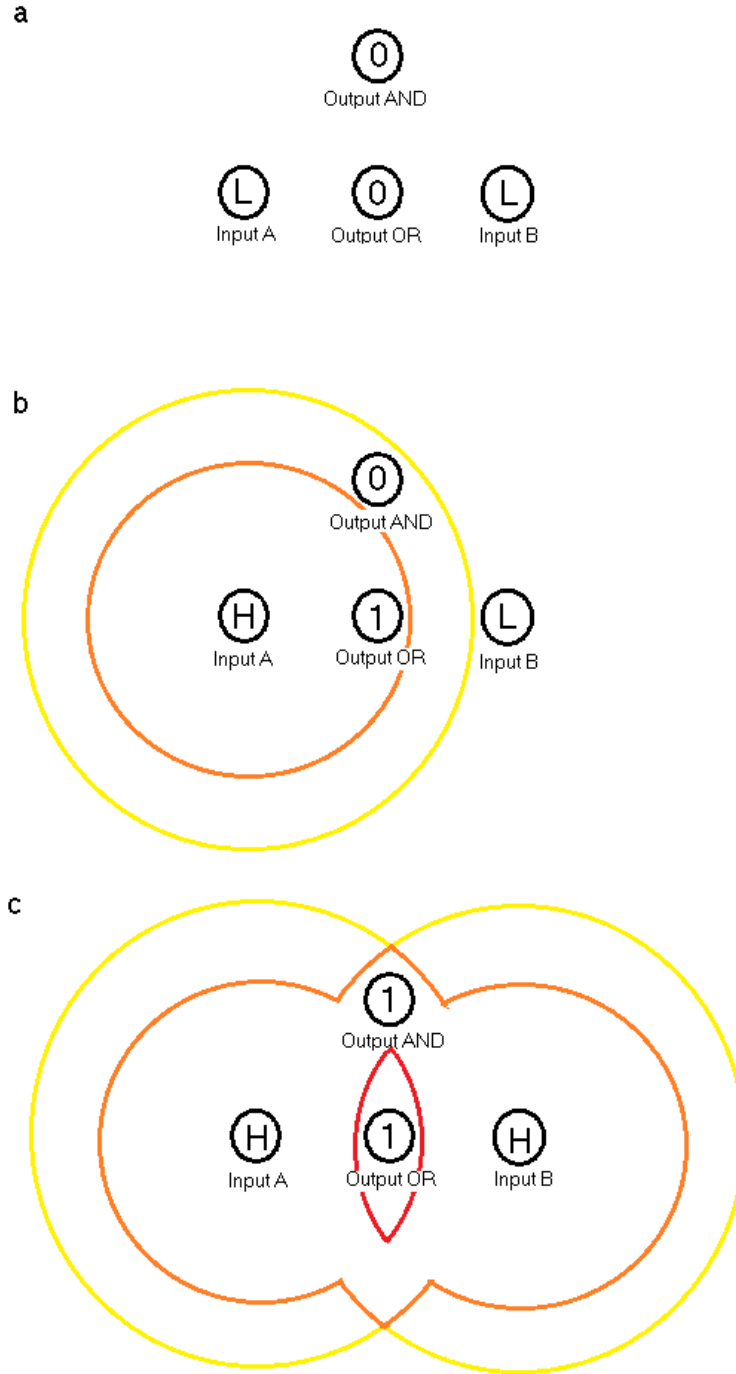


Fig. 2. The Combo gate. (a) Geometry of the inputs and the two outputs. (b) One input high: the inner field reaches Output OR but not Output AND. (c) Both inputs high: the outer field also reaches Output AND. Reproduced from [1].

2.3 Plasmons in monolayer and bilayer graphene

Doped monolayer graphene supports a transverse-magnetic surface plasmon whose dispersion in the long-wavelength limit follows $\omega(q) \propto \sqrt{(q \cdot E_F)}$, where E_F is the Fermi energy set by gate voltage [6, 7, 9]. At $E_F \approx 0.4$ eV the plasmon resonance lies near 10 THz with wavelengths of tens of nanometres, giving an effective field confinement $\lambda_p/\lambda_0 \sim 10^{-3}$. Tuning the gate voltage past the CNP suppresses the carrier density and effectively switches the plasmon mode off. This single mechanism — on/off switching of a propagating bit by gate-voltage-controlled CNP positioning — is the foundation of the present design.

2.4 On-chip terahertz launching

Yoshioka et al. [3] demonstrated that a CMOS-compatible terahertz antenna can launch ultrashort plasmon wavepackets directly into a graphene channel without an external free-space terahertz source. The launched wavepacket has a duration of order 1 ps and a centre frequency tunable from 1 THz to above 10 THz. This experiment removes the historical obstacle to plasmonic logic — the absence of a compact, integrable input stage — and is the experimental anchor of the present design.

With these three ingredients in hand — the Combo-gate construction inherited from [1], graphene plasmons as switchable bits, and an on-chip terahertz launcher in the geometry of Yoshioka et al. [3] — we are now in a position to specify a complete device. AND and OR are already supplied for free by the Combo-gate construction of Section 2.2 once the propagation medium is a switchable graphene channel: the OR output cell sits close to the inputs, and the AND output cell sits farther downstream, both within the same patterned lattice. The only remaining primitive that the architecture must supply explicitly is NOT.

An earlier iteration of this programme [2] realised NOT inside a trilaminar AA/AB-stacked graphene substrate, by draining a negative electrode in the AA region into a positive electrode in the AB region. That route is electrostatically rich but geometrically heavy; here we deliberately adopt a simpler dual-gated bilayer alternative. Two graphene layers are separated by an hBN spacer; the upper layer carries the signal plasmon, and the lower layer is biased independently to launch a π -shifted plasmon. Where the two are routed to coincide, destructive interference performs the NOT operation. The remainder of the paper develops this device concretely.

3. Conceptual model of the THz-clocked half adder

3.1 Device architecture

The device, shown in Fig. 3, modifies the original Yoshioka geometry [3] (Fig. 3a) by adding a second graphene layer, a middle hBN spacer, and an Au bottom gate on the sapphire substrate (Fig. 3b). The top gate (V_{Au1}) and bottom gate (V_{Au2}) are each divided into a lattice of individually addressable segments — in the prototype layout we adopt the $12 \times 6 = 72$ -segment array shown in Fig. 3c — driven by an external electrode-mapping controller. The terahertz launcher is at the left edge of the upper graphene channel.

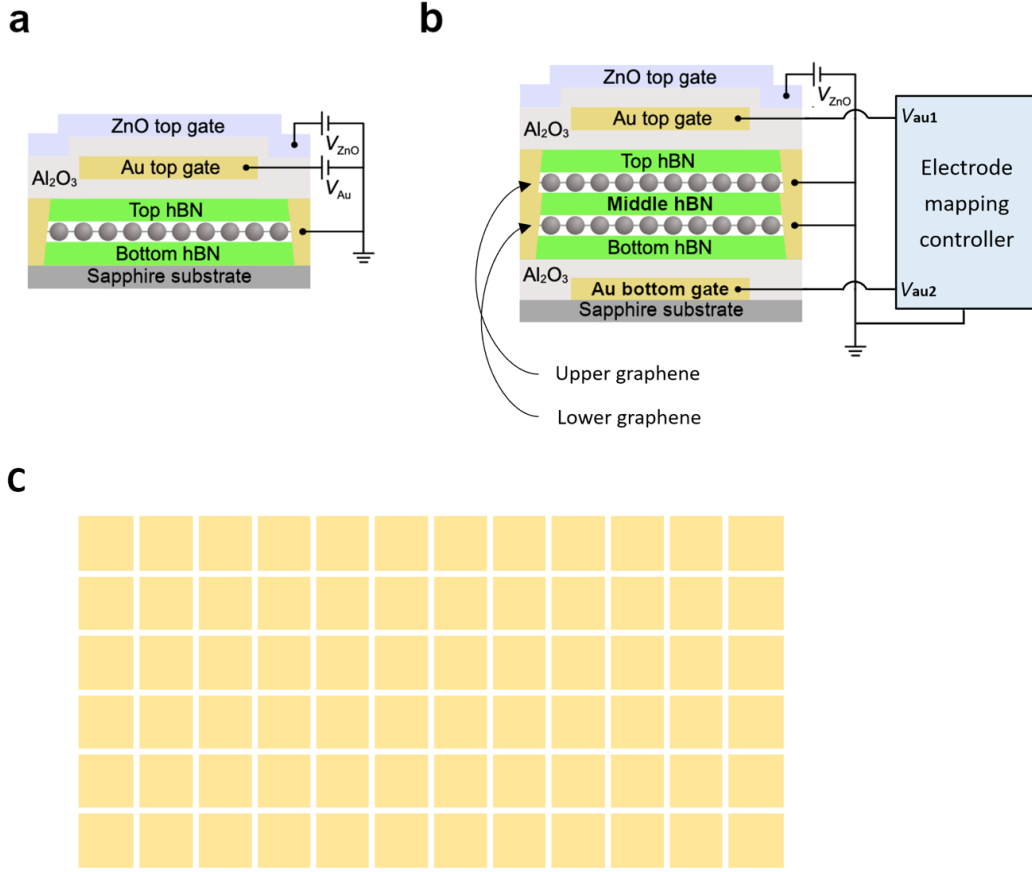


Fig. 3. Cross-sectional view of the half-adder chip. (a) Original device of [3] (single graphene layer between top and bottom hBN, with Au top gate, ZnO top gate, on a sapphire substrate). (b) Modified design with an additional lower graphene layer, a middle hBN spacer, and an Au bottom gate; the two Au gates are driven independently by an electrode mapping controller through V_{Au1} and V_{Au2} . (c) Overhead view of the latticed Au gates (72 segments per gate), each addressable individually.

3.2 Encoding bits as plasmons

We fix the following terminology and use it consistently throughout the paper. A *bit* is the presence (“1”) or absence (“0”) of a plasmon wavepacket of nominal amplitude A_0 at a designated read-out cell at a designated time. The wavepacket itself is launched by a *terahertz pulse* from the on-chip antenna; the pulse and the plasmon are physically distinct, with the pulse being the input drive and the plasmon being the on-chip carrier of the bit. This distinction was implicit in the earlier work [2] and is made explicit here. Read-out is interpreted by an amplitude threshold: at a designated output cell, an integrated near-field intensity above $0.5 A_0^2$ is decoded as 1, below as 0. Section 5 quantifies the noise margin.

3.3 Gate-mapping primitives

On the upper graphene the gate voltage V_{Au1} is set per segment to one of two levels: $V_{Au1} = -1.2$ V parks the local Fermi level at the CNP and blocks plasmon transport, while $V_{Au1} = 0$ V dopes the layer and enables propagation. The set of segments held at 0 V therefore defines the *active path* on the chip (Fig. 5a).

On the lower graphene V_{Au2} is held at one of two levels, -2.4 V or -1.2 V. At -2.4 V the bottom layer is doped sufficiently to support its own plasmon mode; phase control of the terahertz drive ensures that this mode arrives at the interaction node π -shifted with respect to the signal-layer plasmon, giving destructive interference and therefore the NOT operation (Fig. 5b). At -1.2 V the bottom layer is at its CNP and is silent. The required phase tolerance is quantified in Section 5.3.

4. Half-adder pattern and logical verification

4.1 Sum and Carry decomposition

A half adder produces, from two input bits A and B, the Sum = $A \oplus B$ and the Carry = $A \cdot B$. The standard textbook circuit (Fig. 4) realises this with one OR, two AND gates, and one NOT, exploiting the identity $\text{Sum} = (A + B) \cdot \text{NOT}(A \cdot B)$. The dual-gated bilayer of Section 3 supplies all four of these primitives, so the half adder is laid out by tiling them onto the 72-segment lattice as a single Combo-gate region.

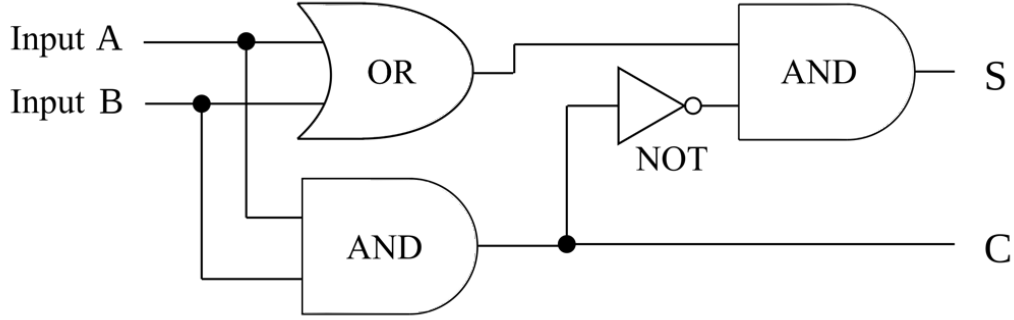


Fig. 4. Traditional half-adder logic circuit (AND, OR, NOT). S = Sum, C = Carry. Reproduced from the present author's earlier manuscript on this device.

4.2 Voltage-controlled gate map

Figures 5a and 5b show, respectively, the upper-gate (V_{Au1}) and lower-gate (V_{Au2}) voltage maps for the half-adder layout. Orange cells in Fig. 5a are at 0 V (active for plasmon transport) and grey cells are at the CNP (-1.2 V, silent). The active orange cells route the inputs A and B (left edge) to the OR junction in the centre, then onward to the AND output S (top right) and to the Carry output C (bottom right), passing through the NOT interaction node. In Fig. 5b, red cells in the lower gate are at -2.4 V and generate the π -shifted inverter plasmon used at the NOT node; the rest of the lower gate is held at -1.2 V (silent).

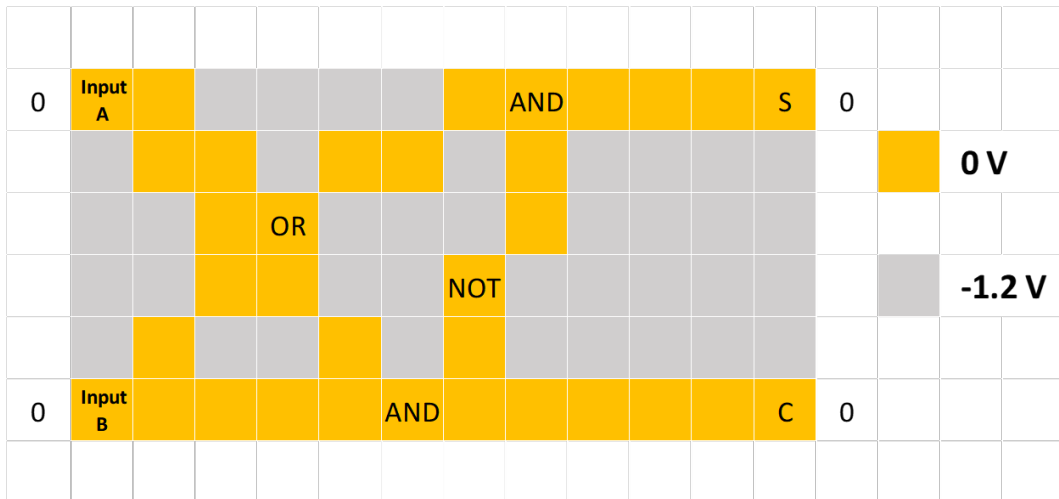


Fig. 5a. Upper Au gate (V_{Au1}) voltage map. Orange = 0 V (active path); grey = -1.2 V (CNP, silent). Inputs A and B enter from the left; outputs S (Sum) and C (Carry) leave at the right.

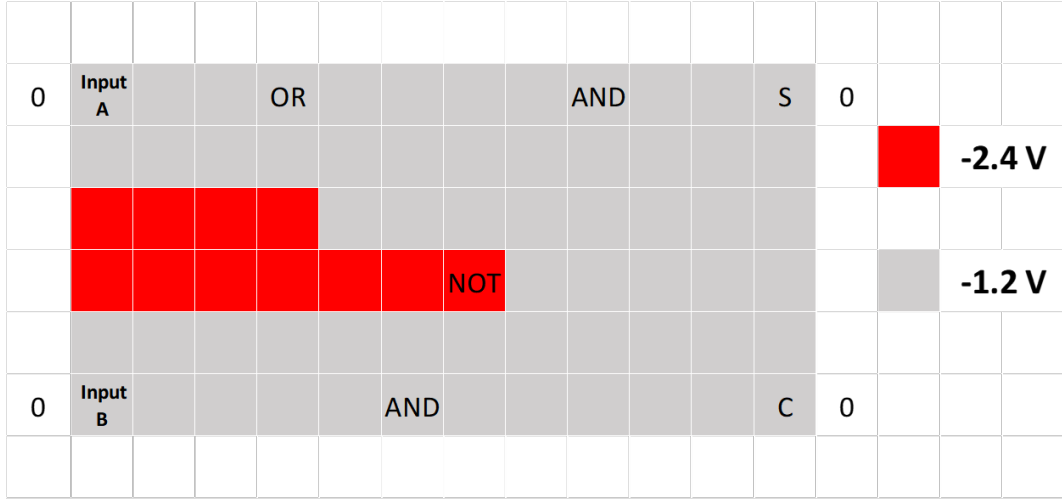


Fig. 5b. Lower Au gate (V_{Au2}) voltage map. Red = -2.4 V (active inverter region, generates the π -shifted plasmon used at the NOT node); grey = -1.2 V (silent).

4.3 Truth-table verification

Figures 6–8 show the device operating in the three non-trivial input combinations ($A=1, B=0$), ($A=0, B=1$) and ($A=1, B=1$). For each case we display the upper-gate plasmon trajectories (panel a) and the corresponding lower-gate inverter pattern (panel b). Plasmon paths are shown as blue arrows.

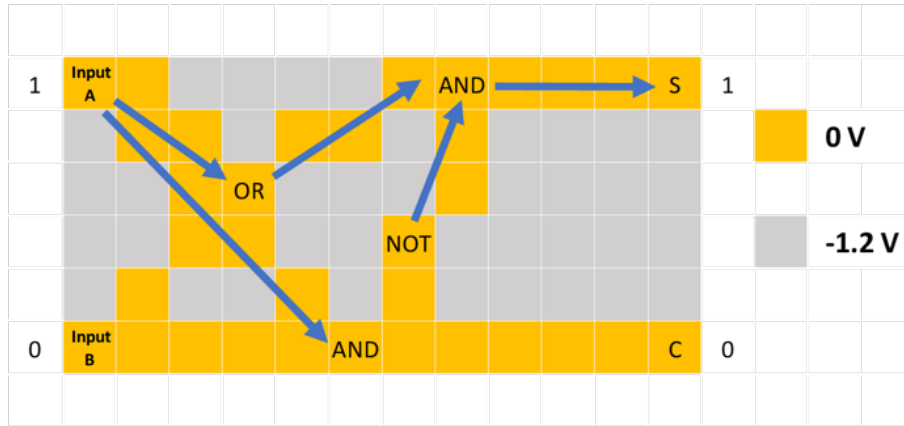


Fig. 6a. Upper-gate plasmon trajectories for $A=1, B=0$. Input A propagates through the OR junction and the NOT-conditioned AND, yielding $S=1$; the bottom AND path requires both inputs and yields $C=0$.

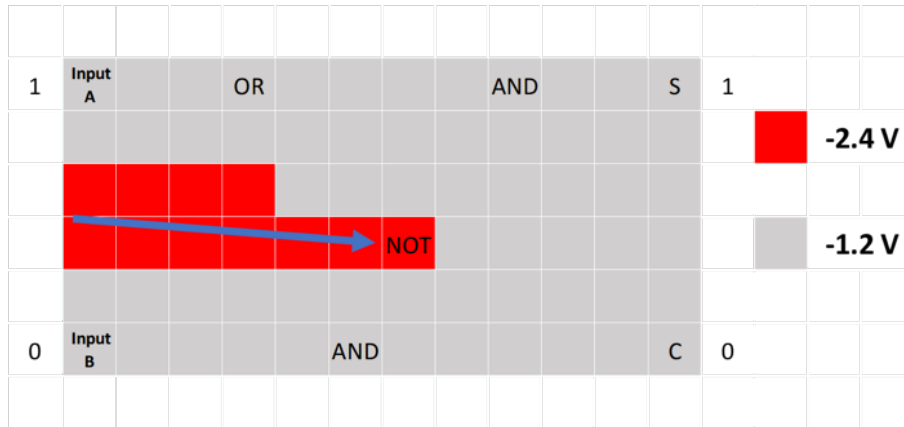


Fig. 6b. Lower-gate inverter pattern for $A=1, B=0$. The π -shifted plasmon is launched only into the NOT region (red = -2.4 V).

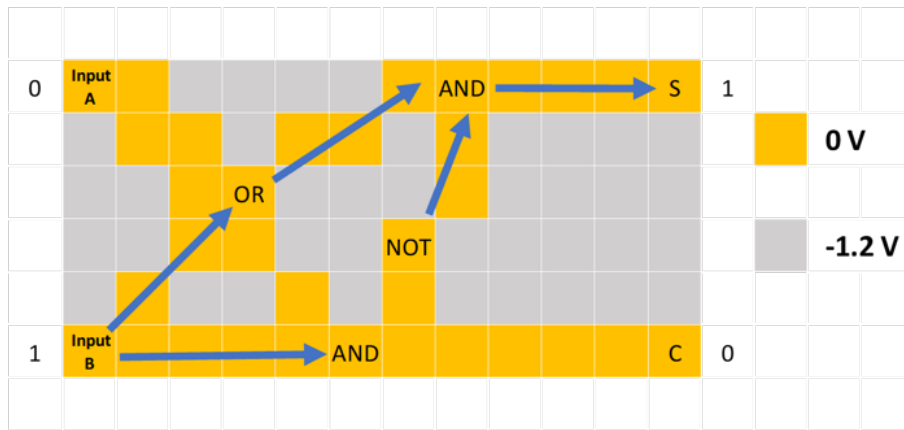


Fig. 7a. Upper-gate plasmon trajectories for $A=0$, $B=1$. By symmetry with Fig. 6a, $S=1$ and $C=0$.



Fig. 7b. Lower-gate inverter pattern for $A=0$, $B=1$; identical to Fig. 6b.

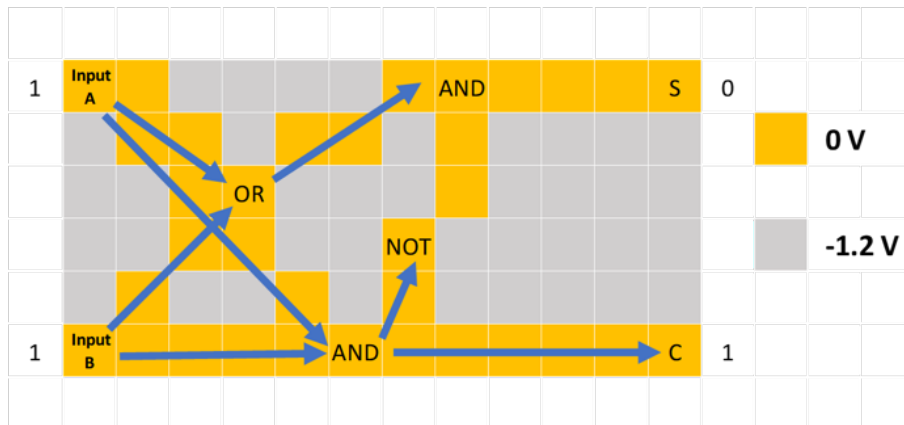


Fig. 8a. Upper-gate plasmon trajectories for $A=1$, $B=1$. Both inputs reach the bottom AND node, producing $C=1$; the OR junction fires and propagates toward the upper AND, but the NOT operation on the AND-of-inputs cancels the Sum output, giving $S=0$.



Fig. 8b. Lower-gate inverter pattern for $A=1$, $B=1$. The inverter region drives the π -shifted plasmon into the *NOT* node, performing the cancellation that suppresses *S*.

The fourth row of the truth table — the ($A=0$, $B=0$) baseline that was absent from the precursor manuscript — is supplied by construction: with no input plasmon launched, no plasmon reaches either output cell and both Sum and Carry read 0. The lower-gate inverter pattern is irrelevant in this case because there is no signal to invert. The full predicted truth table is collected in Table 1.

A	B	Sum ($A \oplus B$)	Carry ($A \cdot B$)	Verified by
0	0	0	0	by construction (no input plasmon)
0	1	1	0	Fig. 7
1	0	1	0	Fig. 6
1	1	0	1	Fig. 8 (NOT cancellation)

Table 1. Predicted half-adder truth table, with each row linked to the figure that supports it. The (0,0) row is supplied by construction; the (1,1) row depends on the inverter-layer *NOT* operating within the phase budget of Section 5.3.

5. Quantitative figures of merit

5.1 Plasmon propagation length and gate delay

Adopting graphene plasmon parameters representative of high-mobility samples on hBN at low temperature [6–8], at a centre frequency of 10 THz and Fermi energy $E_F \approx 0.4$ eV the group velocity is $v_g \approx 5 \times 10^5$ m/s and the plasmon lifetime $\tau \approx 0.5$ –1 ps. The propagation length is therefore $L_p = v_g \cdot \tau \approx 0.25$ –0.5 μm per intrinsic decay length, or roughly 1–3 μm before the amplitude drops below the read-out threshold. The expected gate delay is $\tau_g \approx L_{\text{cell}}/v_g \approx 0.1$ ps for a $L_{\text{cell}} = 50$ nm gate cell, giving an intrinsic clock ceiling above 1 THz — three orders of magnitude above scaled CMOS.

5.2 Energy per bit

An order-of-magnitude estimate of the energy associated with one plasmon wavepacket follows from the volume occupied by the field, $V \sim \lambda_p^2 \cdot d$, with the energy density $(\epsilon/2)|E|^2$ evaluated at the field amplitude needed to register a bit reliably. For the parameters above this gives $E_{\text{bit}} \sim 10^{-18}$ J, or roughly 1 aJ per operation — comparable to single-flux-quantum superconducting logic [10] and several orders of magnitude below scaled CMOS ($\sim 10^{-15}$ J/op). **This is the *intrinsic device* figure: it counts only the field energy stored in the propagating plasmon and excludes the input-antenna conversion, the receive-antenna detection, and the room-temperature drive electronics that surround the chip.** A realistic system-level energy budget will be larger, with the dominant overhead set by the efficiency of the terahertz launcher of [3] and the receive-side detector discussed in Section 6.3; the 1 aJ/op figure should be read as the asymptotic lower bound that this overhead approaches once the I/O stages mature.

5.3 Phase tolerance for the NOT operation

The NOT operation relies on destructive interference of two plasmons of equal amplitude. Writing the residual amplitude as $A_{\text{res}} = A_0 \cdot |1 + e^{i\Delta\phi}|/2 = A_0 \cdot |\cos(\Delta\phi/2)|$, the residual intensity is $I_{\text{res}}/I_0 = \cos^2(\Delta\phi/2)$. Requiring $> 90\%$ extinction ($I_{\text{res}}/I_0 < 0.1$) gives $|\Delta\phi| < \pi/12$, i.e. a tolerance of $\pm 15^\circ$. At 10 THz this corresponds to a path-length tolerance of $\pm \lambda_p/24 \approx \pm 2$ nm and a timing-jitter tolerance of ± 4 fs. Both are within reach of contemporary e-beam lithography and femtosecond-stable terahertz drive electronics, but they set a hard specification on the fabrication tolerance. **For comparison, contemporary phase-locked nonlinear-waveguide systems already operate at a residual phase fluctuation of ~ 9 mrad in the work of Hirota et al. [14], roughly thirty times tighter than the present budget;** the precision regime required by our NOT primitive is therefore comfortably below what has already been achieved experimentally in a comparable nonlinear-waveguide platform (see also Section 6.5).

5.4 Comparison with CMOS and RSFQ baselines

Technology	Clock	Energy per op	Maturity
Scaled CMOS (3 nm)	~ 5 GHz	$\sim 10^{-15}$ J	Mass production
Single-flux-quantum (SFQ) RSFQ	~ 50 GHz	$\sim 10^{-19}$ J	Lab demonstrations
Photonic logic (Si)	~ 100 GHz	$\sim 10^{-14}$ J	Research
This work (THz plasmonic, projected)	> 1 THz	$\sim 10^{-18}$ J	Conceptual / single-gate

Table 2. Order-of-magnitude comparison of the present proposal against representative classical-logic technologies. SFQ figures from [10]; CMOS and Si photonic figures from [4, 5].

6. Discussion: limitations and paths forward

The model developed above is conceptual: it specifies an architecture, verifies it against the full half-adder truth table, and supports it with first-order numbers. Several aspects must, however, be engaged before an experimental demonstration. Rather than catalogue these as a static list of obstacles, we pair each limitation in this section with a concrete research direction that, in our view, addresses it. Where relevant we draw direct analogies with three recent results in nonlinear and terahertz photonics: the demonstration by Yanagimoto et al. [13] of a programmable nonlinear photonic waveguide, which establishes that the operating paradigm we propose for the plasmonic substrate (real-time, lithography-free reconfiguration of a 2D function map under feedback control) is experimentally tractable today; the demonstration by Hirota et al. [14] of 10.1-dB squeezed light from a broadband PPLN waveguide optical parametric amplifier, which provides quantitative and topological precedent for the kind of phase-locking precision and separate-detection-arm topology required by the device's NOT primitive; and the demonstration by Kusyak et al. [15] of a fully monolithic on-chip terahertz circuit with integrated generator, detectors, and in situ reference fabricated on the same sapphire substrate we adopt here, which closes the read-out gap that has historically been the hardest device-level challenge for plasmonic logic.

6.1 Toward room-temperature operation

The plasmon lifetimes used in Section 5.1 are characteristic of hBN-encapsulated graphene at < 100 K [8]. At 300 K phonon scattering reduces τ by roughly an order of magnitude, shrinking L_p to approximately $0.1\text{--}0.3\ \mu\text{m}$ — comparable to the size of a single 50 nm gate cell — and tightening the phase-tolerance budget of Section 5.3. The most promising route to room temperature is full hBN encapsulation of the graphene channel together with the use of *acoustic* graphene plasmons [11], which couple weakly to the optical-phonon channels that dominate room-temperature damping. A complementary route is to operate at higher Fermi energy ($E_F > 0.6$ eV), pushing the plasmon dispersion above the optical-phonon energy and suppressing the dominant decay channel. Either route should be benchmarked first as a single-gate demonstration before being applied to the full half-adder.

6.2 Cascading and amplitude restoration

The single-gate analysis of Sections 4–5 does not yet establish that the plasmon arriving at an output cell can drive a subsequent gate. A workable cascaded chip will require one of three approaches: (i) a gain mechanism in graphene under intense terahertz drive, of the kind discussed in [12], that can amplify the plasmon mid-cascade;

(ii) a transducer-based re-launch stage in which the output of one gate is detected electrically, amplified, and re-launched into the next gate by a local antenna — analogous to the electrical re-clocking stage of an SFQ pipeline; or (iii) tight integration of multiple gates within a single plasmon coherence length, which removes the need for amplification at the cost of gate count per channel. Approach (iii) is the cheapest experimentally and is the natural target for a first multi-gate demonstration.

6.3 Integrated read-out

Detecting the Sum and Carry plasmons as distinguishable electrical signals is non-trivial, and the standard demonstrations rely on near-field optical probing [8] that does not scale to many output cells. A matched terahertz receive antenna at each output cell, operated as a detector counterpart to the launching antenna of [3], is the natural next step. Crucially, this is no longer a speculative requirement. Kusyak et al. [15] have very recently demonstrated a fully monolithic on-chip terahertz optoelectronic circuit on the same sapphire substrate adopted in the present architecture (Section 3.1 and Fig. 3): a generator and two symmetrically placed detector photoconductive switches (PCSs), fabricated in a single lithographic pass from electron-beam-evaporated amorphous silicon, share a single coplanar stripline geometry and operate over the 0.05–1.4 THz band with full galvanic isolation between generation and detection. Their AC-coupled centre-launched design enforces pure odd-mode propagation and reaches a -3 dB point at 440 GHz; the two detector PCSs, placed on opposite sides of the generator, provide simultaneous in situ measurement of signal and reference under identical experimental conditions.

The translation to our half-adder is direct. The terahertz launcher of [3] at the left edge of the upper graphene channel is replaced by an AC-coupled g-PCS in the geometry of [15]; one matched d-PCS is placed at the Sum output cell and a second at the Carry output cell, with their photocurrents read by transimpedance amplifiers and standard lock-in detection as in [15]. The gate-controlled graphene channel of our half-adder occupies the position of Kusyak et al.’s planar sample slot between the two metal traces of the stripline, with the upper and lower Au gates of Section 3.1 doubling as the stripline metal traces — a configuration the authors of [15] explicitly call out under the heading of *stripline gating* as a downstream application of their platform. A third symmetrically placed d-PCS, on a logic-load-free reference row, supplies the in situ reference signal that drives the per-segment trim loop of Section 6.5. Polarisation matching, in-band noise figure, and the geometry of the receive antenna at our 10 THz centre frequency — well above the upper edge of the band characterised in [15] — remain open design problems; but the qualitative gap that previously made integrated read-out the hardest device-level challenge has now been closed by an experimentally demonstrated, monolithic, lithographically reproducible launcher + detector + reference platform built on the same substrate stack we already require.

Each of the three remaining problems admits a concrete design strategy drawn from the existing graphene-plasmonics and photoconductive-antenna literatures. *Polarisation matching* is most efficiently addressed by inserting a graphene plasmonic patch or strip antenna between the photoconductive switch and the logic channel, tuned through its chemical potential to the plasmon frequency and oriented so that the induced current is aligned with the dominant in-plane plasmon field [16, 17, 20]; cross- or dual-polarised graphene-antenna arrays [16] additionally provide per-element phase control if the net field at the channel must be reconfigurable. For the *antenna geometry* itself, the relevant regime above 1 THz sits between conventional photoconductive antennas and pure nano-plasmonics: graphene microstrip and patch antennas tunable from below 1 to roughly 10 THz by gate-controlled chemical potential [20] are directly applicable, while bow-tie or log-periodic shapes scaled to 10-THz dimensions provide ultra-broadband response, and slot antennas integrated into the stripline let the same Au metallisation that gates the graphene also act as the radiating/receiving structure. A leaky- or travelling-wave antenna patterned along the upper-graphene channel is a natural fit for the field-coupled-plasmon picture and can be impedance-matched to the propagating mode without any additional metallisation.

For *noise figure and link efficiency*, the chain is most usefully framed as a link-budget product (launcher efficiency $\times$ coupling to graphene $\times$ propagation loss $\times$ receive efficiency $\times$ detector NEP). Recent reviews of high-performance terahertz photoconductive antennas (PCAs) [18, 19] supply the dominant design levers: PCA material and geometry choices that maximise radiation efficiency, plasmonic nano-structures placed around the PCA gap to enhance the local field, and an explicit complex-impedance match between the PCA, the stripline, and the graphene plasmonic load. The third symmetric d-PCS introduced above is itself a balanced detection arm that

suppresses common-mode noise and lowers the effective noise figure of the chain. A concrete first geometry worth simulating is a bow-tie PCA feeding a graphene patch antenna at 10 THz, with the patch tuned in situ via V_{Au1} to co-resonate with the working-channel plasmon; we leave that simulation to follow-on work, but note that with this combination the three open items collapse into one explicit engineering problem with quantitative knobs (patch dimensions, gate chemical potential, PCA gap geometry, slot length) and a single figure of merit (the link-budget product).

6.4 Refining the NOT primitive

As noted in [2], the trilaminar NOT primitive of the precursor design is the least mature of the three gates. The dual-gated-bilayer NOT used in Section 4 is an evolution of that primitive, but its strict π -phase requirement remains its weakest point. Three variants are worth pursuing in parallel: (i) a Mach–Zehnder-style splitter geometry in which the signal-layer plasmon is divided and one branch passes through a phase-shifting cell, relaxing the requirement of a continuously-driven inverter mode — this variant is topologically aligned with the separate-phase-reference-arm approach of Hirota et al. [14] discussed in Section 6.5, in which a dedicated parallel arm carries the phase-detection function so that detection sensitivity does not have to be traded against losses in the working signal path; (ii) an asymmetric gate-bias scheme that uses small-amplitude offsets to broaden the cancellation null; and (iii) a hybrid in which NOT is performed electrically rather than plasmonically, by reading out the AND-of-inputs and inverting the read-out signal in the receive-antenna stage of Section 6.3. Experimental verification of any of these variants remains open.

6.5 Robustness through real-time feedback calibration

The ± 2 nm path-length tolerance derived in Section 5.3 is achievable per cell but accumulates across many cells, and the CNP voltage of each segment will shift with charged-impurity density during fabrication. A chip-scale layout therefore needs either tight per-segment trim of V_{Au1} and V_{Au2} or, more practically, a calibration loop that adjusts these voltages while the device is operating. A direct precedent for this paradigm has just been demonstrated in a closely related platform: Yanagimoto et al. [13] showed that a programmable nonlinear photonic waveguide can be operated under real-time feedback control, in which the device’s instantaneous behaviour (in their case the second-harmonic-generation efficiency) is measured continuously and used to update the spatial pattern that programs the device, compensating for fluctuations in pump wavelength and environmental conditions. The plasmonic substrate proposed here is well-suited to the same approach: the electrode-mapping controller already addresses each of the 72 segments individually, so a loop that perturbs each V_{Au} setting and reads the resulting change in output amplitude can null both fabrication offsets and slow drift without any change to the underlying device.

A complementary — and quantitatively striking — precedent is provided by Hirota et al. [14], who demonstrated 10.1-dB squeezed light from a broadband PPLN waveguide optical parametric amplifier by reducing the residual phase fluctuation of the phase-locking loop from 14 mrad to 9 mrad. They achieve this by replacing the conventional scheme — in which a fraction of the working signal is tapped to generate the phase-error signal, with the well-known trade-off between detection signal-to-noise ratio and signal-path loss — with a separate phase-detection OPA seeded by tapped probe and pump light *upstream* of the squeezer, so that the working signal itself is never sampled. Two implications are directly relevant to our device. Quantitatively, their 9-mrad fluctuation is roughly thirty times tighter than the $\pm\pi/12 \approx 260$ -mrad tolerance required for $> 90\%$ extinction at the NOT node (Section 5.3); the phase-precision regime needed for our architecture is therefore comfortably within what has already been achieved in a comparable nonlinear-waveguide platform. Topologically, their separation of the phase-detection function onto a parallel arm — specifically so that detection sensitivity can be raised independently of working-signal losses — is the same architectural lesson invoked in Section 6.4 for the Mach–Zehnder NOT variant, and is the one we recommend adopting from the start of any experimental implementation.

We note that the Hirota topology is realised in PPLN, where the phase-detection arm is a second nonlinear waveguide pumped by a tap of the same probe and pump beams. The graphene-plasmon analogue is geometrically simpler: the existing 12×6 lattice already contains many cells whose voltage map can be repurposed at fabrication time as a parallel reference arm. Concretely, one row of the lattice can be reserved as a phase-monitor

channel in which both V_{Au1} and V_{Au2} are driven exactly as in the working channel but with no logic load, and the resulting near-field intensity at a dedicated read-out cell is used as the error signal for the segment-by-segment trim of the working rows. This adds one or two rows ($\sim 15\%$) to the lattice area but requires no change to the fabrication process or to the launcher and receive-antenna stages, and it isolates the phase-locking loop entirely from the working signal, exactly as in the Hirota architecture.

6.6 From a half adder toward programmable plasmonic logic

The most powerful consequence of the architecture is that *the same physical lattice* can compute different logic functions purely by changing the electrode-voltage pattern. A first multi-gate target is a full adder (Sum, Carry, and a Carry-in), followed by an n -bit ripple adder; both can be laid out on the same 12×6 lattice with no re-fabrication. The longer-term aim is a plasmonic field-programmable gate array (FPGA), in which the electrode-mapping controller plays the role of the configuration bitstream. This is a direct conceptual analogue of the lithography-free programmable nonlinear photonic device of Yanagimoto et al. [13], in which a single SiN waveguide is reconfigured into many distinct $\chi^{(2)}$ functions by patterning its optical drive on a photoconductor layer; their experimental demonstration that one physical device can sustain many functions is, in our view, the strongest available evidence that the equivalent shift from one-plasmonic-device-one-function to one-substrate-many-functions is achievable. A second extension worth pursuing is multi-output gates that exploit the two-dimensional character of plasmon propagation — for example, computing Sum and Carry within a single Combo region rather than as separate AND/OR/XOR sub-gates.

6.7 Hybrid digital–analog and neuromorphic operation

The 2024 appendix to [2] anticipated a regime in which the plasmon amplitude at the output cell is not quantised to $\{0, 1\}$ but is treated as a continuous variable. In that regime the same device acts as a multiply-accumulate (MAC) unit: each gate cell becomes a programmable weight, the input plasmons are activations, and the integrated near-field intensity at the output is the inner product. This is a natural substrate for analog neural-network inference. A concrete recipe for on-chip learning is provided by the in situ inverse-design protocol of Yanagimoto et al. [13], in which a QPM grating is iteratively reshaped to maximise an output figure of merit; the plasmonic analogue is a Hebbian-style update of the gate-voltage map driven by the integrated read-out signal at the output cells. Such a learning system would simultaneously exploit the device’s reconfigurability, terahertz clock, and aJ-class energy budget, and is the most attractive long-term application of the architecture. We note that this trajectory — from a discrete logic primitive toward an analog computational substrate operated under feedback — mirrors the parallel emergence of analog optical quantum-computing platforms in adjacent fields, which Hirota et al. [14] explicitly identify as a downstream application of their broadband squeezed-light source.

6.8 A cryogenic classical co-processor for quantum control

Even if full room-temperature operation remains out of reach, a cryogenic version of this device would be valuable as a classical co-processor sitting at 4 K alongside a superconducting quantum processor — removing the I/O bottleneck imposed by today’s room-temperature classical control electronics. The plasmonic substrate is naturally cryogenic-friendly and dissipates orders of magnitude less power than CMOS at the same clock; integrating it with bias-tee terahertz I/O could let the quantum-processor and classical-controller subsystems share a single dilution-refrigerator stage.

7. Conclusions

We have presented a conceptual model of a terahertz-clocked half adder in which the bit is a graphene plasmon wavepacket and the gate vocabulary (AND, OR, NOT, Combo) is inherited from a six-year line of prior work on field-coupled and plasmonic logic. The design has been verified against the complete four-row half-adder truth table, supplied with first-order quantitative estimates for propagation length, gate delay, and energy per bit, and equipped with an explicit phase-tolerance budget for the NOT operation. The known difficulties of the present proposal — cryogenic operation, cascading, integrated read-out, phase coherence at scale, fabrication uniformity, and the maturity of the NOT primitive — have been discussed in Section 6, where each is paired with a concrete research direction. Three recent results in nonlinear and terahertz photonics give us particular confidence in the

operating paradigm: Yanagimoto et al. [13] have shown that real-time, lithography-free reconfiguration of a programmable nonlinear waveguide under feedback control is experimentally tractable today; Hirota et al. [14] have shown that the kind of phase-locking precision (~ 9 mrad) needed for our NOT primitive is already available in a comparable nonlinear-waveguide platform, with a topology — a separate phase-detection arm seeded upstream of the main device — that is well-matched to a Mach–Zehnder NOT implementation; and Kusyak et al. [15] have demonstrated a fully monolithic on-chip terahertz launcher + detector + reference platform on the same sapphire substrate adopted here, removing the principal experimental gap in our read-out chain. We hope this presentation is sufficient to invite simulation, fabrication, and experimental tests by the graphene-plasmonics and terahertz-electronics communities.

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